

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R,S) / S3 (WP) (R,S) / S1 (PT) (S,FE) Examination November 2024 (2019 Scheme)

Course Code: EET201**Course Name: CIRCUITS AND NETWORKS**

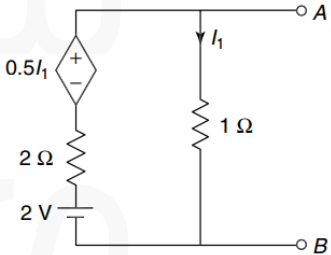
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 State and explain Maximum Power transfer theorem. Also derive the condition for maximum power transfer for DC circuits. (3)

- 2  (3)

Find Norton's Equivalent of the circuit shown above.

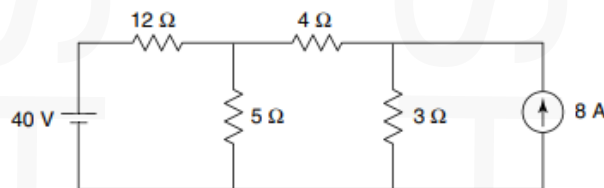
- 3 Write the equations and plot the current responses of RL and RC circuits with DC excitation. (3)
- 4 A 30Ω resistance is connected to a $15H$ inductor in series. A $60V$ DC source is connected to the above-mentioned combination at $t=0$ s. Find the current response $i(t)$. (3)
- 5 State 'Dot rule' for coupled circuits. (3)
- 6 What is a transfer function? What are its four possible forms (3)
- 7 Explain the condition for resonance in a series RLC circuit? What is Quality factor? (3)
- 8 Determine the resonant frequency, bandwidth and Q factor for a series resonant circuit having $R = 10\Omega$, $L = 0.1H$, and $C = 10\mu F$. (3)
- 9 Derive the condition for reciprocity of Z parameters. (3)
- 10 Prove that the resultant Z parameter matrix for the series -connected networks is the sum of Z matrices of each individual two-port network. (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

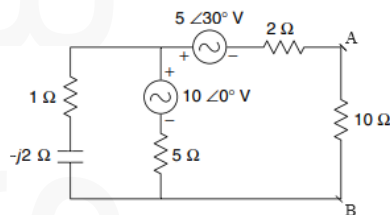
Module 1

- 11a Find Current through 4 Ohm resistor using superposition theorem (12)



- 11b Also find the power dissipated across 4 Ω resistor in the above circuit. (2)

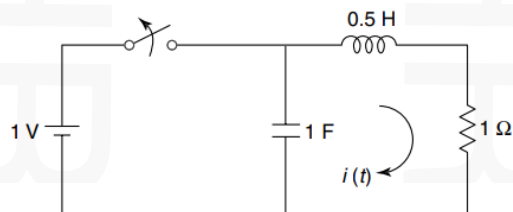
- 12a Determine the Thevenin's Equivalent across terminals A and B (10)



- 12b Calculate the current through 10 Ω resistor in the above circuit. (4)

Module 2

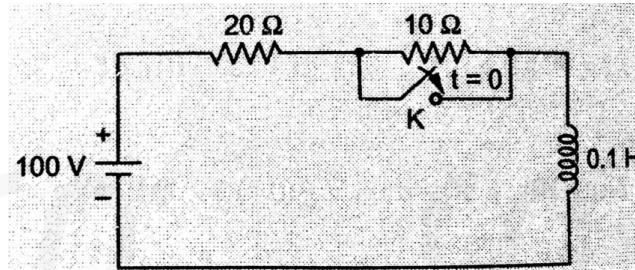
- 13a For the network shown in figure, switch is opened at $t=0$. Steady state condition is achieved before $t=0$. Find $i(t)$ (10)



- 13b Explain the classification of series RLC circuits based on damping ratio. (4)

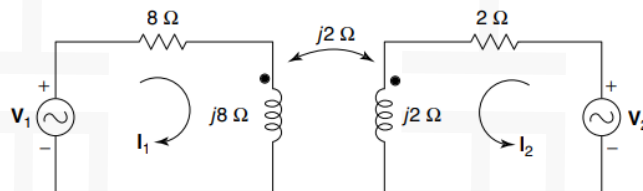
- 14a A sinusoidal voltage of $10\sin 100t$ is connected in series with a switch and $R = 10\Omega$ and $L = 0.1H$. If the switch is closed at $t = 0$, determine the transient current $i(t)$. (7)

- 14b A DC voltage of 100V is applied in the circuit shown below. The switch was kept open for a long time and closed at $t=0$. Find the expression for resulting current. (7)

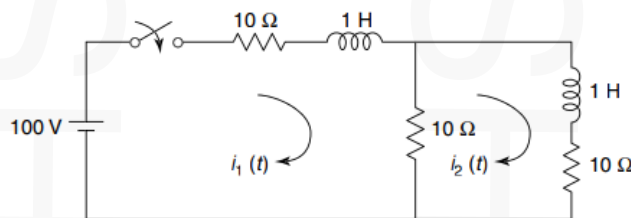


Module 3

- 15a Determine the ratio V_2/V_1 in the circuit if $I_1 = 0$ (10)



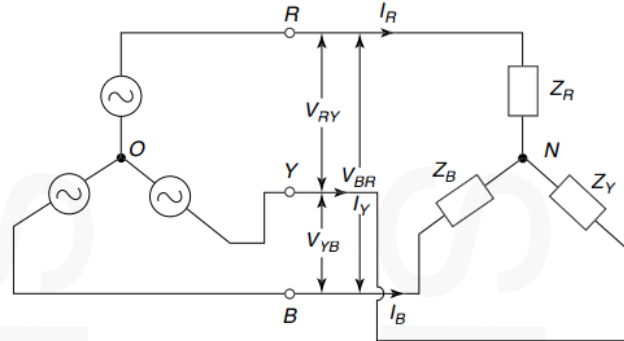
- 15b Draw the S domain equivalent circuit of an Inductor with and without initial conditions (4)
- 16a For the below network, switch is closed at $t=0$. Determine transformed circuit for $t > 0$. Assuming zero initial conditions. (4)



- 16b For the above circuit, determine currents $i_1(t)$ and $i_2(t)$ using mesh analysis (10)

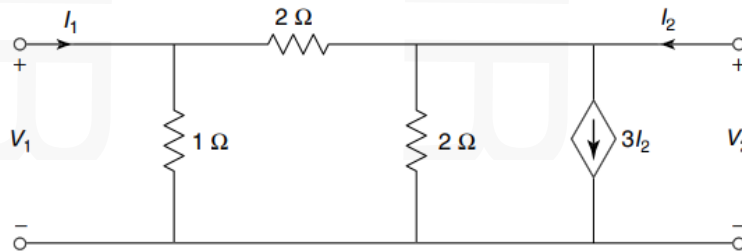
Module 4

- 17a A 400V, 50 Hz, 3 phase supply of phase sequence ABC is supplied to a delta connected load consisting of 100Ω resistance between lines A and B, 318mH inductance between lines B and C and $31.8\mu\text{F}$ capacitance between lines C and A. Determine phase and line currents. (10)
- 17b What is neutral shift in a three-phase unbalanced star connected load. (4)
- 18 A symmetrical 440 V, 3-phase system supplied a star-connected load with the branch impedances $Z_R = 10\Omega$, $Z_Y = j5\Omega$, $Z_B = -j5\Omega$ as shown in figure below. Calculate line currents and voltage across each phase impedance. The phase sequence is RYB. (14)

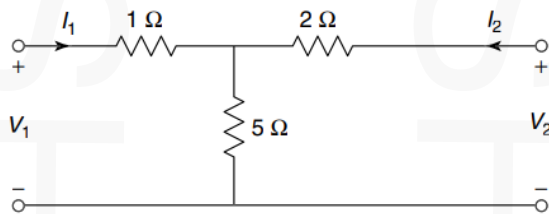


Module 5

- 19 For the network shown below find Z and Y parameters (14)



- 20a Find Transmission parameters for the network (8)



- 20b Derive the expression for h parameters in terms of z parameters. (6)
